

BOEKBESPREKINGS/BOOK REVIEWS

Elements of Sampling Theory

by Vic Barnett

First Edition, xiii + 152 pages, 2 illus., 11 figures and 2 tables; The English University Press Ltd., St Paul's House, Warwick Lane, London, 1974; (U.K. £1.55 paper).

As the author states in his preface, it becomes increasingly impossible to cover all aspects of statistics at the undergraduate level. One such topic is sample survey methods. This monograph is suitable as a basis for an introductory course in this field. The presentation is not at an advanced mathematical level, which renders it suitable for use during the partical sessions of a statistics course. The book certainly deserves a place in the library of research workers using statistical analysis in applied fields such as medicine, psychology, business administration and sociology. It also provides a stepping stone to more advanced reading on the subject of sample survey methods. In this respect the book provides adequate references to more advanced textbooks.

Chapter 1 considers some non-statistical aspects of survey sampling, and sets up the model for probability sampling. Chapter 2 is concerned with simple random sampling as a basis for estimating population means, totals and proportions. In Chapter 3 ratio and regression estimators are discussed. Chapters 4 and 5 deal with simple stratification and clustering methods whilst Chapter 6 provides a brief description of more complicated sampling schemes.

Throughout the book the relative merits of the various estimation methods are discussed and the formulae given are in terms of the finite population correction factor.

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Stochastic Processes and Applications in Biology and Medicine Part I and II

by M Iosifesco and P. Tăutu

First edition in English, 331 pages (part I) and 337 pages, 7 figures and 5 tables (part II). Springer-Verlag, Berlin, 1973. (DM 53, —: US \$21,20 per part.)

This book, published in two parts as volumes 3 and 4 in the series Biomathematics (edited by K. Kirckeborg, R.C. Lewontin, J. Neyman and M. Schreiber), is an expanded and improved version of the authors' book which appeared in Rumanean in 1968. According to the authors the book is intended to introduce mathematicians and biologists

with a strong mathematical and probabilistic background to the study of stochastic processes and their applications in the biological sciences.

Part I deals with the theory of stochastic processes, more specifically with those parts of the theory which are the most relevant to biomedical applications. It consists of two chapters. In the first chapter discrete-parameter stochastic processes are dealt with, the emphasis being laid on denumerable Markov chains. The second chapter contains the theory of continuous-parameter stochastic processes. In particular processes with independent increments and Markov jump and diffusion processes are treated. The mathematical treatment of the theory is clear and rigorous and this part is suitable as a textbook for a course on Markov chains and Markov processes on third year under graduate or first year post graduate level.

In part II important stochastic models in demography, genetics, epidemics, chronic diseases, physiology and pharmacology are surveyed. Each model is described in detail from both the biological and the probabilistic points of view. Special reference is also made to recent developments. This part, illustrating the application of Markov processes in an applied field, is useful to the lecturer lecturing in stochastic processes, as well as to the post graduate student taking a course therein.

Both parts of the book contain an extensive bibliography, which comprises 35 pages in part I and 44 pages in part II. A sound knowledge of probability theory is required to understand the material presented. The mathematical level, however, is too high to be of any value to biologists in South Africa with very few exceptions if any. For the biometrician it is a very useful reference book and is indispensable on his bookshelf.

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Asymptotic Theory of Rank Tests for Independence

by F.H. Ruymgaart

First edition, 115 pages. Mathematical Centre Tracts 43, Mathematisch Centrum, Amsterdam, 1973. (Dfl 12.00)

In the preface to the monograph the author summarizes the main results: "The approach introduced by Chernoff and Savage to prove asymptotic normality under fixed alternatives of rank statistics for the two-sample problem has first been applied, in a suitably modified form, by Bhuchongkul to prove asymptotic normality of rank statistics for testing the hypothesis of independence. In the present tract this method is generalized so that weaker conditions on the limiting score function suffice for the asymptotic normality under fixed continuous bivariate distribution functions, belonging to the alternative. The condition frequently encountered in the literature that the limiting score function, which is a function on the unit square in the independence problem, be a product of functions on the unit interval is abandoned. Apart from this secondary generalization, the main results lie in an essential enlargement of the orders of magnitude of the limiting score function near the boundary of the unit square, and in the fact that this function may exhibit discontinuities on a finite number of lines in the unit square, parallel to the axes. Discontinuity of the limiting score function entails, however, a local differentiability condition on the underlying bivariate distribution function."

In chapter 1 the problem of independence is stated and some locally most powerful rank tests are introduced. In chapter 2 asymptotic normality under fixed alternative distribution functions is obtained for limiting score functions that are continuous throughout the open unit square. A similar result is established in chapter 3 for limiting score functions that are allowed to have discontinuities of a special kind. In chapter 4 the results of the preceding chapters are applied to prove asymptotic normality under converging sequences of alternatives. Applications to consistency and asymptotic relative efficiency are also given in this chapter. The verification of the conditions of these theorems in some important special cases is dealt with in chapter 5. Finally, in chapter 6 attention is paid to discrete underlying alternative distribution functions, entirely concentrated on a finite lattice of points in the plane.

Readers who are acquainted with the Chernoff-Savage technique of proving asymptotic normality of rank statistics will appreciate a number of interesting applications of this method. The notation used is complicated but clear enough and the exposition fair. The typewritten text is not too well spaced.

The author succeeded in his aims as stated above. This monograph is written for the specialist and deserves a place on the book-shelf of every research worker in nonparametric statistics.

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L'Analyse des Données

J.P. Benzécri et Collaborateurs

First edition in two volumes. Volume 1 : La Taxinomie viii+615 pages Volume 2 : L'Analyse des Correspondances vii+619 pages
Dunod, Paris, 1973. (fr 224)

These two volumes, published in French, constitute a methodical presentation of 10 years of research by staff and students in Prof. Benzécri's statistical laboratory at the Faculty of Science in Paris. Each volume has the same general structure in four parts : an introductory section summarising the concepts and methods to be presented; a second section comprising a number of "lessons" which treat in detail the mathematical aspects of the statistical techniques used; a third section presenting a series of completed studies in a wide spectrum of applied fields such as botany, zoology, ecology, psychology, linguistics, economics and politics; and finally a fourth section containing computer programs of the various techniques used.

The main feature common to all the material presented is the almost supreme importance of the data in the development of a statistical technique and, correspondingly, the reduced emphasis of the researcher's ideas or hypotheses "a priori" in determining both the direction taken by an analysis and the degree of structure introduced into the multivariate populations being studied. In short, the model must follow the data, not the inverse. This approach implies a certain criticism of many methods commonly used in practice but it is beyond the scope of this review to enter into a discussion of what is, after all, a fundamental difference in the philosophy of data analysis.

The first volume deals principally with different techniques of clustering. After studying various definitions of similarity and distance in a multivariate context and introducing the various algorithms of clustering in the literature, the fundamental notions of hierarchical generalised algorithm is given for the construction of an "ascending hierarchy" (i.e. when the classification proceeds by successive aggregations of the individuals or classes of individuals). The choice of the distance and the clustering criterion depend on the nature of the problem and the data and these are included as options in the flow of the algorithm. Information is naturally lost in the process of clustering and a separate chapter in which various notions of information theory are developed demonstrates the optimality in this respect of the minimum variance approach using the so-called chi-square distance. Here too there is the added advantage of a link between the clustering analysis and correspondence analysis (treated in volume 2) and it is possible to recognize the descriptive variables responsible for the formation of a cluster of individuals. Other methods of clustering are treated in detail, for example, those which rely on successive partitioning of the individuals, thus forming a "descending hierarchy", those optimising a partition already constructed by an algorithm of exchange between classes, and finally certain algorithms of rapid clustering when treating high dimensional data.

Volume 2 deals with correspondence analysis, a variant of principal component analysis with certain desirable properties, for example the symmetric role played by the individuals and the attribute variables and subsequently their simultaneous representation on the coordinate axes (hence the term "correspondence"). After a general chapter on the euclidean representation of a set of points provided with masses and distances in high dimensional space, the chi-square distance function is discussed in detail along with algebraic and geometric aspects of this metric structure. Correspondence analysis is thus formally defined as the simultaneous search for the principal axes of inertia of two sets of points (defined by the rows and the columns) associated with a contingency table. The same solution is obtained in the more general framework of the product space of the two finite sets where the symmetry is more clearly demonstrated, and the whole argument for two finite sets can be generalised to the case of two infinite spaces. The "factors" derived from different matrix forms — matrices containing logical elements (e.g. ones and zeros only, symmetric matrices, etc. — are investigated, and various subsidiary aspects are treated such as the validity of a factor and polynomial relationships among the factors.

Throughout the second volume matrix and vector notation with its strict asymmetric row and column aspect is rejected in favour of the isomorphic notation of linear mappings. This notation demonstrates without ambiguity the notions of quadratic forms and vector and dual spaces inherent in the theory. Also included in the text are numerous examples derived from examination papers along with their solutions.

The work as a whole is lengthy since it tends to be too complete and a much shorter version is conceivable. The studies presented in detail in the third section of each volume are all of interest but there is a fair amount of repetition because each is written by a different author. Otherwise these two volumes, in their scope and in their presentation, certainly represent a unique publication in the field of applied statistics.